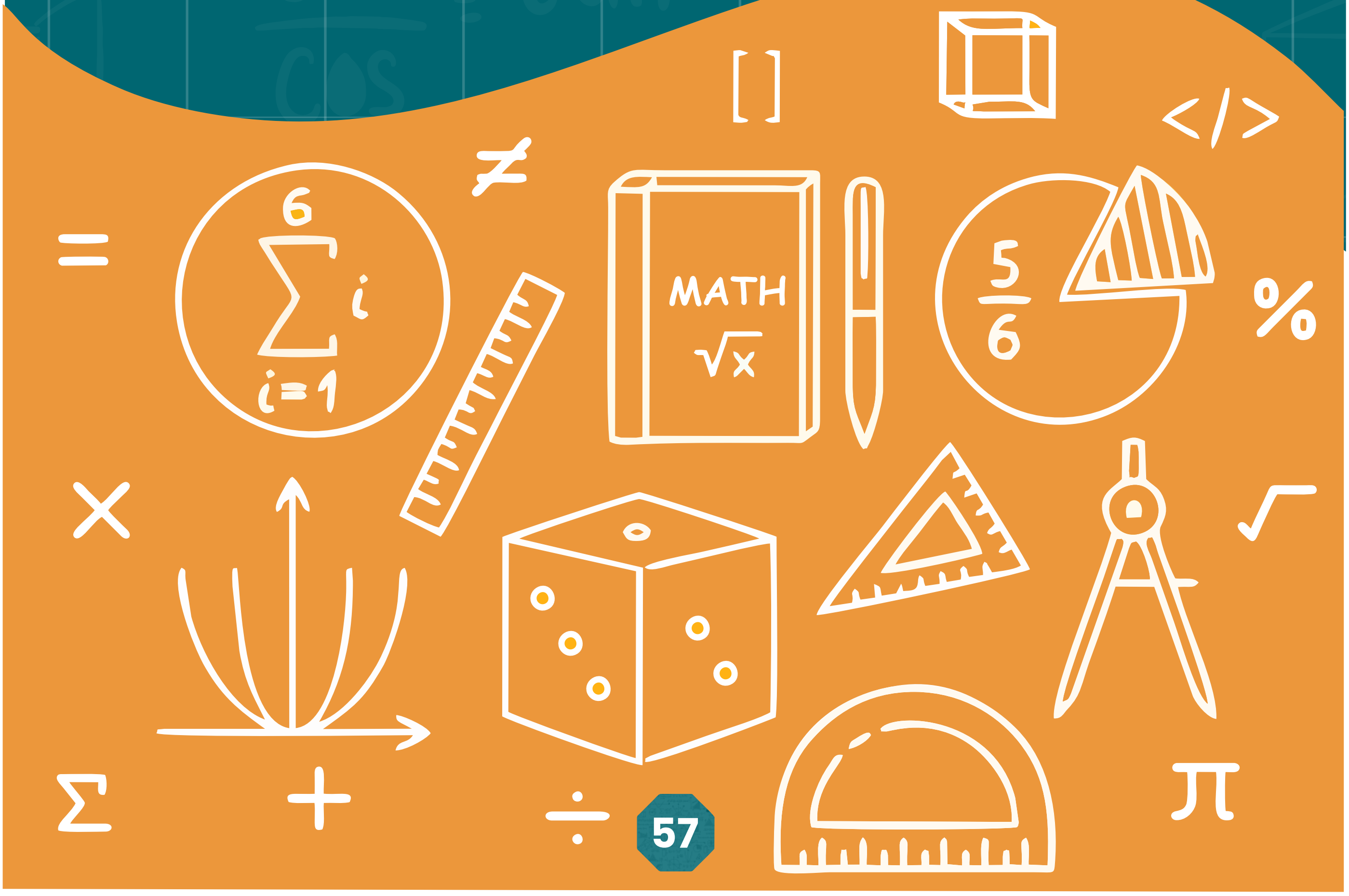
UNIT 5

POLYNOMIAL



POLYNOMIAL

ALGEBRAIC EXPRESSION

A combination of variables and constant obtained by algebraic operations of addition (+), subtraction (-), multiplication (x), division (/), root extraction (√) and raising to powers, is called an algebraic expression or simply an expression.

1. For example:

(i) x **(ii)** -5 **(iii)** xyz **(iv)** $\sqrt{x}$ **(v)** $\frac{1}{2}$ x 2 $\frac{1}{3}$ x $\frac{1}{5}$ x $\frac{1}{4}$ are algebraic expression.

Constant:

An entity having a fixed numerical value is called a constant. For example in expression 2x, 3x + 9y – 1, 2, 3, 9 and -1 are constants respectively.

Variable:

A symbol or letter used to represent some element of a non-empty set is called a variable or unknown. Variables are denoted by small (lower case) letters x, y, z, etc. of the English alphabet.

Literal:

A letter used to represent a constant or a variable in an expression is known as literal. For example in ax + by + c, a, b, and c are literals representing constants.

Coefficient:

A coefficient is a constant which is a multiplier of the variable in a term.

POLYNOMIALS

A polynomial is an algebraic expression involving a sum of positive integer powers in one or more variables multiplied by coefficients. A general form of a polynomial is:

$$P(x) = a^n x^n + a^{n-1} x^{n-1} + \ldots\ldots + a^1 x^1 + a^0$$

where:

- a^n , a^{n-1} , ..., a^1, a^0 are constants called coefficients.
- x is the variable.
- n is the degree of the polynomial.

5.2.1 Types of Polynomials

Polynomials can be categorized based on the degree and number of terms:

- **Monomial:** A polynomial with one term.

Example: $5x^3$

- **Binomial:** A polynomial with two terms.

Example: $x^2 - 4$

- **Trinomial:** A polynomial with three terms.

Example: $3x^2 + 2x - 1$

POLYNOMIAL

Degree Of Polynomial

The degree of a polynomial is the highest power of the variable in the polynomial.

Example:

• $3x^4 + 2x^2 - 5$ has degree 4.

• $7x - 1$ has degree 1.

• $xy + y - 1$ has degree 2. (Because the total degree of the term xy is 1 + 1 = 2 which is the highest degree.

Standard Form of a Polynomial

A polynomial is written in standard form when the terms are arranged in descending order of the powers of the variable.

Example:

$$5x^3 + 3x^2 - 2x + 7$$

EXERCISE 5.1

Q1. Separate constant, variable and literals from each of the following algebraic expressions.

1. $3x + 1$
2. 0
3. $2a^2 + \frac{1}{3}b - 3$
4. -8
5. $Px + qy + r$
6. $-5x + 9y - 4$
7. $3l - 2m$
8. $y^2 - y - 1$
9. $a + b^2 - 2c + 4$

Q2. Identify polynomials from the following:

1. -1
2. $\frac{3}{y}$
3. $5xy^3$
4. $\sqrt{5}\,x + y$
5. $\frac{1}{x} - x$
6. $ax^2 - bx + c$

Q3. Indicate the degree of the polynomials :

1. 3
2. $x^2y^2z^2 + 1$
3. x
4. xy^2
5. $3^2\,xyz$
6. $-2\,xyz$
7. $\frac{2}{13}$
8. 4^2
9. $-x^3 + 8xyz^2$

Q4. Write coefficients of the following polynomials.

1. 3
2. $-\frac{3}{4}x^2$
3. $\sqrt{4}\,xy$
4. $-x$
5. $\sqrt{2}\,x^3 - \sqrt{3}\,y^4$
6. $5x^3 + 9y + z$

OPERATIONS ON POLYNOMIALS

Operations like addition, subtraction, multiplication, and division can be performed on polynomials.

• **Addition and Subtraction:** Combine like terms.

• **Multiplication:** Multiply each term of one polynomial by every term of the other polynomial.

• **Division:** Polynomial division is similar to long division.

Example1. **Add $2x + 3y - 4z$ and $5z + 6x - 3y$**

Solution:

Vertical Method

$$\begin{array}{r} 2x + 3y - 4z \\ +\quad 6x - 3y + 5z \\ \hline 8x - 0y + 1z \\ = 8x + z \end{array}$$

Horizontal Method

$= (2x + 3y - 4z) + (5z + 6x - 3y)$

$= 2x + 3y - 4z + 5z + 6x - 3y$

$= 2x + 6x + 3y - 3y + 5z - 4z$

$= 8x - 0y + 1z \;\; = 8x + z$

Example2. **Add $2a + 3b + 4$ and $6 + 5a - 6b$**

Solution:

Vertical Method

$$\begin{array}{r} 2a + 3b + 4 \\ +\quad 5a - 6b + 6 \\ \hline 7a - 3b + 10 \end{array}$$

Horizontal Method

$= (2a + 3b + 4) + (6 + 5a - 3y)$

$= 2a + 3b + 4 + 6 + 5a - 6b$

$= 2a + 5a + 3b - 6b + 4 + 6$

$= 7a - 3b + 10$

Example 3. **Multiply $x^2 - 2x - 5$ by $x + 3$**

Solution:

$= (x^2 - 2x - 5)(x + 3)$

$= x^2(x + 3) - 2x(x + 3) - 5(x + 3)$

$= x^3 + 3x^2 - 2x^2 - 6x - 5x - 15$

$= x^3 + x^2 - 11x - 15$

EXERCISE 5.2

Q1. Find the sum of given polynomial:

1. $P(x) = 2x^2 + 3x - 5$ and $Q(x) = x^2 - 4x + 7$.

2. $P(x) = 4x^3 + x^2 - 2x + 3$ and $Q(x) = -2x^3 + 5x^2 - 3x - 1$

3. $P(x) = 5x^2 - 3x + 2$ and $Q(x) = -x^2 + 4x - 6$.

4. $P(x) = x^3 + 2x^2 - x + 5$ and $Q(x) = -x^3 + 3x^2 + 4x - 2$

5. $P(x) = 7x^3 - 4x^2 + x - 1$ and $Q(x) = -3x^3 + 5x^2 - 2x + 4$

Q2. Find the difference $P(x) - Q(x)$:

1. $P(x) = 3x^2 - 5x + 7$ and $Q(x) = x^2 + 2x - 4$

2. $P(x) = x^3 + 2x^2 - 3x + 5$ and $Q(x) = -x^3 + 4x^2 + 2x - 1$

3. $P(x) = 4x^2 - 7x + 3$ and $Q(x) = 2x^2 + 3x - 1$

4. $P(x) = 6x^2 - 4x + 2$ and $Q(x) = x^2 + 3x - 4$

5. $P(x) = 7x^2 - 6x + 8$ and $Q(x) = 5x^2 - 4x + 2$

Q3. Multiply:

1. $P(x) = x + 3$ and $Q(x) = x - 2$

2. $P(x) = x^2 + 4x + 3$ and $Q(x) = x + 1$

3. $P(x) = x^2 - 2x + 3$ and $Q(x) = x - 1$

4. $P(x) = x - 3$ and $Q(x) = x^2 + 5x + 6$